

[DC] Enhancing Accessibility and Emotional Expression in Educational Extended Reality for Deaf and Hard of Hearing: A User-centric Investigation

Sunday D. Ubur*

Virginia Tech

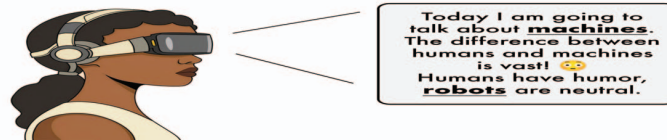


Figure 1: A user reading real-time transcription with highlighted keywords, including emojis to convey emotional expression.

ABSTRACT

My research focuses on addressing accessibility challenges for the Deaf and Hard of Hearing (DHH) population, specifically within Extended Reality (XR) applications for education. Emphasizing improved communication, the study aims to enhance accessibility and emotional expression in XR environments. Objectives include identifying best practices and guidelines for designing accessible XR, exploring prescriptive and descriptive aspects of accessibility design, and integrating emotional aspects for DHH users. The three-phase research plan includes a literature review, development of an accessible XR system, and user studies. Anticipated contributions involve enhanced support for DHH individuals in XR and a deeper understanding of human-computer interaction aspects.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Accessibility—Accessibility technologies; Social and professional topics—User characteristics—People with disabilities

1 INTRODUCTION AND MOTIVATION

Accessibility for the Deaf and Hard of Hearing (DHH) population is a critical issue that requires attention to ensure equitable access to education, healthcare, and other essential services. With a focus on education, the inclusion of DHH students in mainstream classrooms remains a controversial topic, requiring strategies to better support their academic achievements [2, 8]. This controversy arises from the challenges DHH students face in mainstream settings, such as limited access to information and communication barriers.

One important accessible technological solution for the DHH is Speech-to-Text (STT), which plays a pivotal role in enhancing communication in an inclusive setting. The ability to convert spoken language into text facilitates communication in various contexts, ranging from online meetings to educational settings. However, amidst the strides made in improving accessibility, a critical aspect often overlooked is the incorporation of emotional expression into transcribed text generated by STT systems.

Extended Reality (XR) has shown great potential in various fields, including education [14], for example to visualize transcribed text and reduce cognitive load with embodiment and immersion for the user [4].

*e-mail: uburs@vt.edu

The inclusion of all DHH individuals, including DHH children, in XR applications is still not a reality [12]. This is particularly important in the context of education, where XR has the potential to enhance the learning experience for DHH students, making education accessible to all learners, including DHH students [6, 11].

Further, enhancing emotional expression in STT through XR is essential in order to ensure the emotional impact of XR on learning and learning outcomes [8, 10], especially by enhancing engagement and providing immersive experiences through visual aids. XR has the potential to impact affective and cognitive learning outcomes, indicating its relevance in influencing emotional expression [9].

As the impact of XR technologies in teaching and learning continues to grow both in mainstream and DHH populations [1], especially regarding learning outcomes [8], creating XR environments that support accessibility and emotional expression in STT is essential for ensuring usability, usefulness, emotional impact, and meaningfulness [5] for the DHH user.

The expected contributions of this research include:

- Improved accessibility support for DHH individuals in XR, particularly in educational settings, by providing specific guidelines and practices tailored to their needs.
- A deeper understanding of HCI aspects related to XR accessibility for DHH users, with a focus on user interaction and interface design.

By exploring the intersection of XR, accessibility, emotional expression, and STT technology, this research aims to pave the way for more effective and emotionally engaging communication interfaces.

2 RESEARCH QUESTIONS

In our HCI research lab (Distributed Virtual Environments Lab), as a DHH individual who actively uses XR devices, I am acutely attuned to the prevailing limitations in XR design, particularly its inadequacy in catering to the accessibility needs of the DHH community. The intricacies of communication between virtual and real worlds, exacerbated by the insufficient emotional expression in STT, serve as the impetus for the focal point of my research.

I conducted a narrative literature review that underscored a conspicuous gap in XR accessibility research for the DHH, coupled with a notable absence of guidelines for both embodied user interfaces and emotional expression in STT. My research endeavors to address and remedy these shortcomings. The prevalent deficiencies in XR design contribute to a sense of isolation among DHH individuals, attempting to navigate the dual realms of XR and the physical world, thereby resulting in a sub-optimal user experience and heightened cognitive load.

The specific emotional impacts on DHH individuals, encompassing the challenges of understanding communicated messages and immersing in the XR environment while having stress-free access to transcribed real-world messages through STT, underscore the imperative for enhanced accessibility.

Thus, **the first research question** seeks to explore best practices and guidelines for incorporating accessibility in XR for the DHH in educational contexts: *What are the best practices for DHH XR accessibility in educational contexts?*

A systematic literature review focusing on AR in education reveals a stark scarcity of research on AR for DHH individuals [3]. This accentuates the need for a theoretical model to augment XR accessibility. Another study emphasizes the absence of guidelines for rendering XR content visible to assistive technologies, particularly in addressing auditory disabilities [13].

The second research question aims to comprehend the specific accessibility requirements for DHH individuals in XR, with a focus on proposing prototypes that enhance UI design, fonts, color, and text. Figure 1 illustrates the basic concept.

To further enrich this, an exploration of alternative user interfaces, such as embodied user interfaces [4, 7] can be tailored to meet the needs of DHH users. This motivates the research question: *What are the design guidelines for embodied user interfaces to make immersive educational environments less isolating for DHH?*

The third research question is poised to investigate the integration of emotional expression in STT within the XR environment, considering the unique challenges and experiences of DHH individuals, including sensory overload during XR technology usage: *How to address emotional aspects in DHH XR accessibility design, considering the unique challenges and experiences of DHH individuals, including sensory overload?*

3 EXPECTED NEXT STEPS

Phase 1 — Systematic Literature Review (SLR): In this phase, a comprehensive review will be conducted to understand existing knowledge on XR accessibility for DHH individuals, with a particular focus on emotional expression. The aim is to identify current best practices for affordances and embodied cognition in STT user interfaces and systems.

Phase 2 — Design and Development of DHH XR Accessibility Guidelines: This phase involves developing an XR system with emotional capabilities for DHH individuals, addressing identified gaps. A user-centered design approach will be employed, with active involvement of DHH individuals in the design process. The integration of emotional expression features will consider the feedback obtained in Phase 1.

Phase 3 — Evaluation and User Study: In this phase, the effectiveness and usability of the developed XR accessibility solution will be assessed through user studies with DHH participants. The evaluation will specifically focus on the integration of emotional expression in the XR system, considering its impact on the user experience and emotional well-being of DHH individuals.

Overall Expected Contribution: Better support for DHH XR accessibility and a deeper understanding of the HCI practices, guidelines and theories involved. This will involve engaging in an interdisciplinary exploration at the intersection of XR, accessibility, emotional expression, and STT technology.

4 BENEFITS OF IEEE VR 2024 DOCTORAL CONSORTIUM PARTICIPATION

As a dedicated Ph.D. student in computer science, specializing in accessibility within HCI research, participating in the DC is a valuable opportunity. This platform allows me to share my Ph.D. research with the vibrant HCI community, gaining crucial feedback to enhance its depth and impact.

Engaging with the HCI community through the DC offers a unique chance for constructive feedback from accomplished researchers and peers. This diverse input will refine my research's methodology and scope, providing a broader perspective crucial for navigating my Ph.D. journey.

The insights obtained will shape my work, ensuring it goes beyond academic excellence to make a meaningful impact on society and the field of computer science. The constructive feedback received will contribute significantly to the depth and significance of my research focus.

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